

LAURA BURNETT, PhD

Research Data Scientist | Research Software Engineer

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PROFILE

I build software that makes large scientific datasets usable and informative to a broader audience. Most recently I worked on *neuView*, which turns a connectome database of more than 150,000 cells into a browsable web resource. Before that I ran studies end to end, designing the experiments and writing the code for every stage: stimulus generation, data acquisition, quality control, statistical analysis and the final figures. I move readily between scientific domains, and want to broaden my data science and machine learning skills in a team building tools that matter beyond the lab.

SKILLS

Languages

Python (pandas, NumPy, Plotly, Dash) • MATLAB • SQL • Cypher (Neo4j) • JavaScript, HTML & CSS

Software practice

Git & pull-request workflow • continuous integration (GitHub Actions) • packaging & dependency management (pyproject, pixi) • technical documentation (Quarto)

Data & methods

ETL & pipeline design • statistical modelling & hypothesis testing • time-series & signal processing • image and video analysis • deep learning for automated annotation (DeepLabCut) • dimensionality reduction & clustering • graph and network analysis

Visualisation

interactive web resources • dashboards • SVG & JavaScript visualisation • 3D neuroanatomical visualisation (Neuroglancer) • article figures (Affinity Designer)

Core competencies

reproducible & executable analyses • FAIR data principles • open-source development • mentoring & teaching • communicating technical work to non-specialists

RESEARCH & ENGINEERING EXPERIENCE

Postdoctoral Research Scientist

Reiser Lab, HHMI Janelia Research Campus • 📅 Aug 2023 – May 2026 • 📍 Ashburn, VA, USA

➤ Connectomics & graph databases • *Drosophila* visual neuroscience • high-throughput behavioural screening • interactive web resources • open research hardware • reproducible workflows

- **Built *neuView*, an open, web-based research tool.** A Python application that turns a Neo4j connectome database into a browsable page for every cell type in the fly nervous system, containing pre-computed statistics, interactive graphics and connectivity tables. Scaled it from the visual system alone to all ~12,000 cell types, and designed it to regenerate itself for new datasets as they are released. Selected contributions include the embedded Neuroglancer 3D view with interactive features and the hexagonal occupancy heatmaps extended with JavaScript so they can be explored layer by layer. Tested with pytest, built and deployed through GitHub Actions, packaged and released under GPL-3.0.
- **Developed code to analyse a *Drosophila* connectome — classified ~53,000 neurons into 732 cell types.** One of a core team of 12. Specifically contributed the spatial patterning analysis to understand how every cell type samples the visual field. Published in *Nature* (2025) as an *executable paper*, with every figure reproducible from a public repository. My plotting code was later adopted by an external connectome-analysis package, *connectome_interpreter*.
- **Designed and ran a high-throughput behavioural screen.** Built the entire pipeline: stimuli for the lab's custom LED arenas, acquisition scripts, and synchronisation across camera, stimulus and environmental logging, through to QC, statistical analysis and figure generation. Added dashboards for browsing the data and wrote the public documentation. The workflow outlived the project: the rig and pipeline are now in routine use by new members of the group running their own experiments, and a paper drawing on the screen is in preparation.
- **Wrote research software for a patch-clamp electrophysiology collaboration.** A two-pass adaptive protocol: the first pass maps a cell's receptive field, the second generates stimuli in real time, confined to that region. The documentation I created for this project became a crucial shared resource among the various team members.
- **Tested new open-hardware LED displays for visual neuroscience.** Joined the small team developing the sixth generation of the lab's modular LED arena as its experimental tester, writing the code that verified new panels could render and run visual stimulus patterns. The design drives the arena from a Teensy microcontroller instead of costly data-acquisition hardware, putting open, low-cost visual neuroscience rigs within reach of smaller labs and science communicators.

PhD Researcher — Systems Neuroscience

Joesch Lab, Institute of Science and Technology Austria (ISTA) 📅 Sep 2016 – May 2023 📍 Klosterneuburg, Austria

➤ Systems & behavioural neuroscience · neurodevelopmental conditions and autism models · visual perception & decision-making · *in vivo* electrophysiology · optogenetics · computational behavioural analysis

- **Led a multi-year study from question to publication.** Asked whether genetically distinct mouse models of autism share the same perceptual impairments, and where in the brain they arise. Owned the experimental design and analysis plan, and formed a new collaboration between two research groups to deliver it. First author in *PLoS Biology* (2024). I also built the experimental rigs I worked on.
- **Trained and validated deep-learning pose-estimation models** (*DeepLabCut*) to track pupil position and full-body pose in behaving mice. Designed the annotation strategy, tested how well models generalised across sessions and animals, and built the pipeline converting raw tracking into quantitative behavioural measures.
- **Built analysis pipelines for large, noisy multimodal data** — video, behavioural and electrophysiological time series — from raw signals through to statistical inference. Wrote all of my own analysis code, and developed guidelines and best practices that were adopted by incoming members of the lab.
- **Deliberate breadth:** rotations in bulk RNA-seq analysis, microfluidic device fabrication and two-photon microscope assembly, alongside advanced statistics coursework during my first year.

Earlier research positions

UCL Institute of Child Health · University of Melbourne · The Florey Institute 📅 2014 – 2016

📍 London, UK & Melbourne, Australia

➤ Nervous system development (enteric & cortical) · quantitative analysis of confocal microscopy image data

- Four research placements across three institutions — Master's project with the Burns Lab (UCL ICH), undergraduate placements with the Young Lab (Melbourne) and Tan Lab (Florey), and a graduate summer internship with the Hippenmeyer Lab (ISTA) — contributing to publications in *Neuron*, *Nature Communications* and *Developmental Biology*.

SOFTWARE & OPEN SOURCE

- [neuView](#) — Python application generating a static, interactive cell-type explorer for a Neo4j connectome database of >150,000 cells, with a pytest suite, GitHub Actions CI, packaged, GPL-3.0. Deployed at reiserlab.github.io/celltype-explorer-drosophila-male-cns.
- [reiser-documentation](#) — Documentation site covering experimental protocols, analysis code and rationale for a research lab; written as a reference and for people joining the work.
- [cv-toolkit](#) — Markdown-to-PDF CV toolchain with automatic page fitting. Unit and end-to-end pytest suites, GitHub Actions CI.
- leburnett.github.io — personal portfolio and technical writing, built with Quarto.

TEACHING & MENTORING

- **Supervision and mentoring:** twelve people across two institutions — one research technician, one graduate student, seven graduate rotation students, two summer undergraduates and two high-school summer students (Janelia 2023–2026; ISTA 2016–2023).
- **Teaching Assistant**, Neuroscience Track Core Course, ISTA (2018–2020): taught the image analysis with MATLAB section to incoming first-year graduate students.
- **Summer camp session lead**, ISTA Fakebusters (2021, 2022): designed and led a one-day session, “*Seeing is believing? Vision and optical illusions*”, for ~25 high-school students.
- **Mentoring group member**, Janelia Women and Non-binary Mentoring Group (2023–present, now remote alumni member).

TALKS & PRESENTATIONS

Invited talks: joint symposium with the Champalimaud Centre for the Unknown (2021); joint symposium with Universität für Bodenkultur Wien (BOKU), ISTA (2021).

Posters: NeuroFly, Birmingham (2024); FENS Forum, Paris (2022) and online (2020); FENS International Winter School, Sölden (2022); Austrian Neuroscience Meeting, Salzburg (2021) and Innsbruck (2019); Colour Vision: Circuits and Behaviour, HHMI Janelia (2019).

Internal seminars and lab meetings, ISTA and HHMI Janelia (2016–2026).

PUBLICATIONS

h-index 5 · 408 citations (Google Scholar).

1. Berg S, Beckett IR, Costa M, ... **Burnett LE**, et al. Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*. 2026;189(18):5504–5526.
2. Nern A, Loesche F, Takemura S, **Burnett LE**, et al. Connectome-driven neural inventory of a complete visual system. *Nature*. 2025;641(8065):1225–1237.
3. **Burnett LE**, Koppensteiner P, Symonova O, et al. Shared behavioural impairments in visual perception and place avoidance across different autism models are driven by periaqueductal grey hypoexcitability in *Setd5* haploinsufficient mice. *PLoS Biology*. 2024;22(6):e3002668.
4. Hao MM, Bergner AJ, Nguyen HTH, Dissanayake P, **Burnett LE**, et al. Role of JNK, MEK and adenylyl cyclase signalling in speed and directionality of enteric neural crest-derived cells. *Developmental Biology*. 2019;455(2):362–368.
5. Beattie R, Postiglione MP, **Burnett LE**, Laukoter S, Streicher C, Pauler FM, et al. Mosaic analysis with double markers reveals distinct sequential functions of *Lgl1* in neural stem cells. *Neuron*. 2017;94(3):517–533.
6. McCann CJ, Cooper JE, Natarajan D, Jevans B, **Burnett LE**, Burns AJ, et al. Transplantation of enteric nervous system stem cells rescues nitric oxide synthase deficient mouse colon. *Nature Communications*. 2017;8(1):15937.

EDUCATION

PhD, Neuroscience

[Institute of Science and Technology Austria \(ISTA\)](#) ☞ Sep 2016 – May 2023 📍 Klosterneuburg, Austria

Thesis: *To flee, or not to flee? Using innate defensive behaviours to investigate rapid perceptual decision-making through subcortical circuits in mouse models of autism.*

MSci Natural Sciences — Neuroscience (First Class Honours)

[University College London \(UCL\)](#) ☞ Sep 2015 – May 2016 📍 London, UK

BSc Natural Sciences — Organic Chemistry & Biomedical Sciences (First Class Honours)

[University College London \(UCL\)](#) ☞ Sep 2012 – May 2015 📍 London, UK

[University of Melbourne \(study abroad, Dean's List\)](#) ☞ July 2014 – July 2015 📍 Melbourne, Australia

A-Levels — Mathematics (A*), Biology (A*), Spanish (A*), Chemistry (A)

[King's High School, Warwick](#) ☞ 2010 – 2012 📍 Warwickshire, UK

OUTREACH & COMMUNITY

- **Volunteer maths tutor**, Action Tutoring (2026–present): weekly Year 6 maths tuition for students from disadvantaged socio-economic backgrounds.
- **STEM outreach**, HHMI Janelia (2024–2026): “Hour of Code”, teaching primary-school children to code in Python; RESET / ScienceExplorers, hands-on inquiry-based STEM activities for 5th-grade students.
- **Science communicator**: founding member of *Wissen Schafft's* (2021), a multilingual YouTube series created with ISTA colleagues to tackle misinformation during the COVID-19 pandemic.
- **Founder and organiser**, ISTA Art Club (2018–2023); **Graduate Student Association Representative**, ISTA (2018–2020).
- **Event volunteer**, EX2 Adventures (2024–2026): trail running and mountain bike races in Virginia and Maryland.

FUNDING & AWARDS

Travel grant, Austrian Neuroscience Association, FENS (2022).

OeAD research scholarship — competitive, self-authored proposal (2016).

Dean's List, University of Melbourne (2014) — awarded for outstanding academic achievement, placing in the top 3–5% of the cohort.

REFERENCES

Available on request.